Dahyun Joo

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Academic Background

Seoul National University 2025 – Present

*M.S. Course, Mechanical Engineering, **GPA 4.0/4.0***

- AI-driven Simulation and Design Lab (Principal Investigator: Prof. Do-Nyun Kim)

Seoul National University 2019 – 2025

*B.S. Physics (Minor: Mechanical Engineering), **Cum Laude***

- Thesis: *wPINNs: Enhancing Physics-Informed Neural Networks with Wavelet Activation Functions*
  *Compulsory Military Service (Jul. 2021 – Apr. 2023, Republic of Korea Air Force)

Chungnam Science High School 2016 – 2019

*R&E at Physics Student Group, **Valedictorian***

Research Experience

M.S. Researcher Mar 2025 – Present

SNU AI-driven Simulation and Design Lab (Prof. Do-Nyun Kim)

- Develop **reusable kirigami-based impact absorbers** by engineering geometry-driven **negative-to-positive stiffness transitions** and energy absorption for repeatable deformation without material damage.
- Build simulation-to-design pipelines (e.g., **FEA**, parametric modeling, optimization loops) to map pattern parameters to force–displacement response and impact performance.
- Develop an **explainable ten-qubit quantum neural network** for multi-material topology optimization, with transfer to unseen loading conditions, higher-resolution meshes, and voxel-wise 3D problems.

Undergraduate Research Intern Sep 2024 – Jan 2025

SNU MetaStructure Lab (Prof. J. H. Oh)

- Studied **waves and metamaterials** using acoustic finite element analysis and supported AI-driven topology-optimization simulations and design iterations.

Undergraduate Research Intern Oct 2023 – Jan 2024

SNU Transformative Architecture Laboratory (Prof. J. K. Yang)

- Performed Abaqus FEA of **deployable bistable dodecahedral origami** to assess fabrication feasibility and guide design iteration.

Publications

D. Joo, Y. Miyazawa, S. Hong, D. Lee, J. K. Yang, D.-N. Kim, *Kirigami Meta-Sheet for Enhanced Impact Absorption*. **Submitted;** arXiv:2609.08362 [cs.CE], 2026.

- Introduces a planar, enhanced kirigami impact absorber that dissipates local impacts through distributed deformation and transitions between positive- and negative-stiffness regimes, validated through experiments, finite-element simulations, and reduced-order modeling.

D. Joo, N. Sukulthanasorn, K. Terada, D.-N. Kim, *Explainable Quantum Neural Networks for Multi-Material Topology Optimization*. **Under Review;** arXiv:2607.00438 [cs.CE], 2026.

- Develops a ten-qubit explainable quantum neural network that predicts structural layouts and material assignments from multi-material topology-optimization histories, generalizing to unseen loading conditions, higher-resolution meshes, and voxel-wise 3D problems while retaining auditable observable channels.

Conference Presentations

D. Joo, Y. Miyazawa, S. Hong, D. Lee, J. K. Yang, D.-N. Kim, *Planar Kirigami Impact-Mitigation Sheet with Geometrically Tuned Stiffness Transitions*. **Oral**, KSME Annual Meeting 2026, Nov. 11 – 14, 2026.

D. Joo, N. Sukulthanasorn, K. Terada, D.-N. Kim, *Transferable Multi-Material Topology Prediction with Mechanics-Interpretable Quantum Observables*. **Poster**, 12th Korea Multi-Scale Mechanics Symposium (KMSM 2026), Aug. 13 – 14, 2026.

Projects

Samsung Display 2026.03 – 2027.02

AI-driven Pattern Discovery for Improved Hinge Deformation Behavior

- Built an automated 2D modeling–simulation workflow over unit-cell and hinge geometries, generating AI-training datasets and a surrogate model for hinge-strain prediction and geometry selection.

Samsung Display 2025.03 – 2026.02

Bistable Kirigami Integration for Curved Display Corner

- Integrated and parameterized bistable kirigami patterns for controlled shape transition under corner-curvature, boundary, manufacturability, and repeatability constraints.

LG Electronics 2025.06 – 2025.09

Kirigami-Enhanced Pulp-Mold Packaging for Impact Absorption

- Developed kirigami-enhanced pulp-mold packaging concepts and practical pattern guidelines for improved impact absorption under large-scale forming constraints.

Patents

Samsung Display, D.-N. Kim, **D. Joo**, G. Yun, K. Jung, *Bistable Metapatterns with Dome Geometry and Optimal Design Method*, patent pending.

Awards

STOB League: Semiconductor Solverton Competition, Top Prize 2024

- Deputy Prime Minister and Education Minister Prize.
- Optimized GAA nanosheet FET AC performance using structural modeling and TCAD; analyzed resistance, capacitance, and frequency gain relative to FinFET.

Seoul National University Semiconductor Specialized College Scholarship 2024

Korean History Proficiency Test (Level 1, 100/100) 2022

ROKAF Entrepreneurship Contest: Advanced to MND Finals 2022

Excellent Private Award (ROK Air Force Information Communication School Principal) 2021

Certificate of Completion: Startup Investment Education (SNU Entrepreneurship Center) 2020

Excellent Student Award (Chungcheongnam-do Federation of Teachers' Association) 2019

Seminar

Seminar Instructor (Reinforcement Learning) Jan 2024

SNU BI Lab (Prof. Byoung-Tak Zhang)

- Presented reinforcement-learning concepts and game-theoretic foundations, including minimax decision-making, Nash equilibrium, and links to policy/value learning, for an interdisciplinary audience.

Professional Experience

Intern — K-Startup Center Paris at STATION F 2025.02

Intern — Thales Korea 2024.03 – 2024.07

Working Scholar — SNU Gwanak Residence Halls Administrative Office 2024.01 – 2024.02

Content Creator — `@biztorang` & `@jjuvoyager` 2022.10 – 2023.08

Sergeant (Completed Service) — Republic of Korea Air Force (15th Weather Squadron) 2021.07 – 2023.04

- Developed leadership and operational-coordination skills in a high-responsibility environment.

President of Student Council — SNU Department of Physics and Astronomy 2020.11 – 2021.07

- Led student representation and cross-group coordination.

Working Scholar — Center for Theoretical Physics, SNU Department of Physics and Astronomy 2020.03 – 2021.06

Brand Owner / Online Retail Sales — `jjooda` & `DogodoNongwon` 2020.04 – 2023.05

CFD Engineer — Rocket pre-startup `Kspace` (Injector Design) 2019.12 – 2020.02

Skills

Languages: Korean (Native), English (TOEFL 104; GRE V155/Q170/AW4.0), French (DELF A2)
Programming: C, Python, MATLAB, R | **Programs**: ABAQUS, ANSYS, NX, STAR-CCM+, SolidWorks, COMSOL, KeyShot, Cura
Fabrication / Testing: FDM, SLA and metal 3D printing; ARAMIS 3D Motion and Deformation Sensor, GOM Correlate; laser cutting; UTM; milling